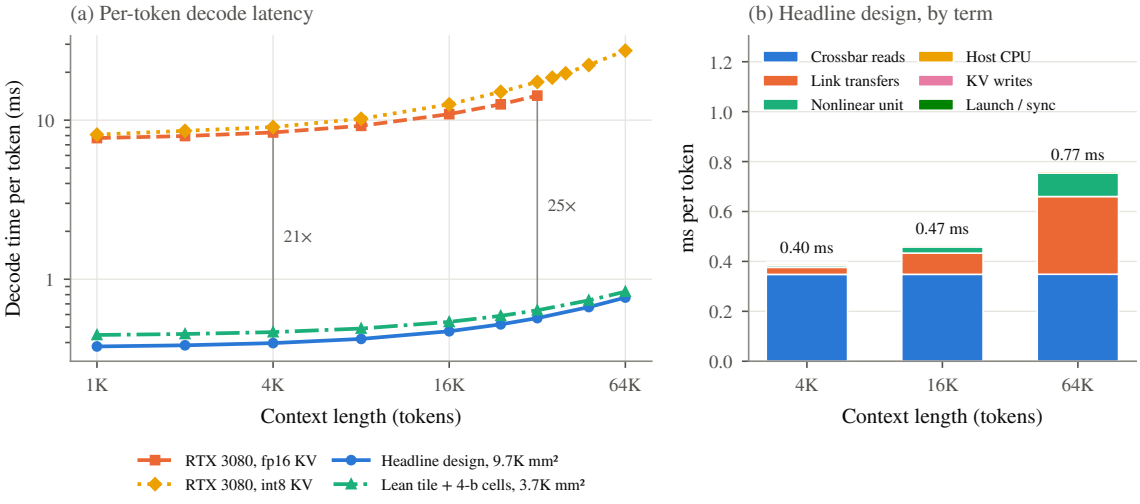
(a) Per-token decode latency
Decode time per token (ms)
10
1
21×
25×
1K
4K
16K
64K
Context length (tokens)
RTX 3080, fp16 KV
RTX 3080, int8 KV
Headline design, 9.7K mm²
Lean tile + 4-b cells, 3.7K mm²
(b) Headline design, by term
Crossbar reads
Link transfers
Nonlinear unit
Host CPU
KV writes
Launch / sync
ms per token
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0.40 ms
0.47 ms
0.77 ms
4K
16K
64K
Context length (tokens)